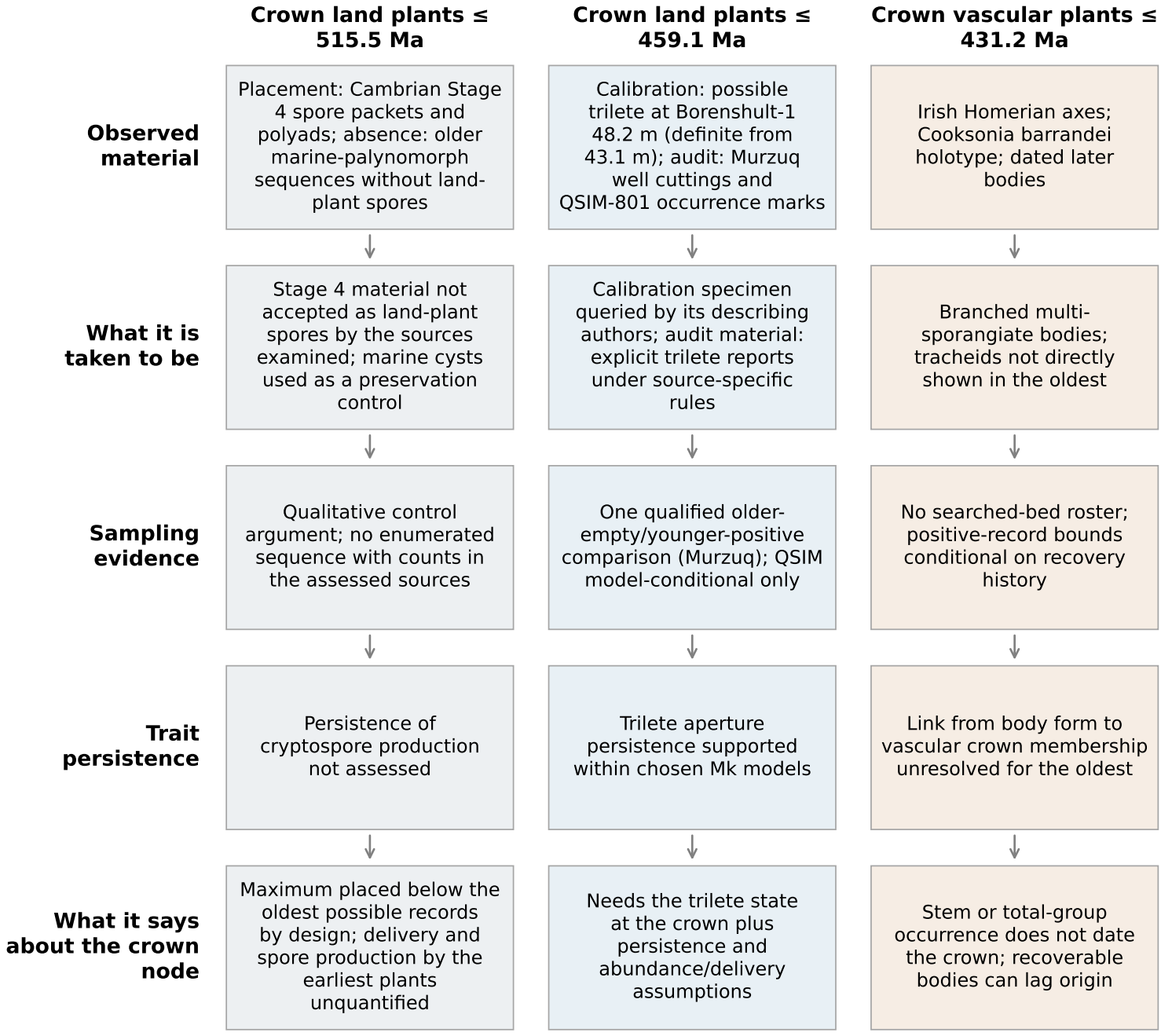
Crown land plants ≤ 515.5 Ma
Crown land plants ≤ 459.1 Ma
Crown vascular plants ≤ 431.2 Ma
Observed material
Placement: Cambrian Stage 4 spore packets and polyads; absence: older marine-palynomorph sequences without land-plant spores
Calibration: possible trilete at Borenshult-1 48.2 m (definite from 43.1 m); audit: Murzuq well cuttings and QSIM-801 occurrence marks
Irish Homerian axes; Cooksonia barrandei holotype; dated later bodies
What it is taken to be
Stage 4 material not accepted as land-plant spores by the sources examined; marine cysts used as a preservation control
Calibration specimen queried by its describing authors; audit material: explicit trilete reports under source-specific rules
Branched multi-sporangiate bodies; tracheids not directly shown in the oldest
Sampling evidence
Qualitative control argument; no enumerated sequence with counts in the assessed sources
One qualified older-empty/younger-positive comparison (Murzuq); QSIM model-conditional only
No searched-bed roster; positive-record bounds conditional on recovery history
Trait persistence
Persistence of cryptospore production not assessed
Trilete aperture persistence supported within chosen Mk models
Link from body form to vascular crown membership unresolved for the oldest
What it says about the crown node
Maximum placed below the oldest possible records by design; delivery and spore production by the earliest plants unquantified
Needs the trilete state at the crown plus persistence and abundance/delivery assumptions
Stem or total-group occurrence does not date the crown; recoverable bodies can lag origin